

Acid-Base Geometry: Understanding Acid and Base Behavior as Charge-Distribution Geometry

A Mechanistic Reframing of Acid-Base Theory from First Principles of Charge Distribution

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Abstract

Acid-base chemistry is taught through three historical frameworks — Arrhenius, Brønsted-Lowry, and Lewis — each of which correctly identifies participants in acid-base reactions but stops short of explaining the physical mechanism driving those reactions. This paper proposes that all three frameworks are downstream consequences of a single underlying physical reality: the spatial distribution of electron density in molecules, here called **charge-distribution geometry**. An acid is a molecular geometry configured to compress its charge distribution upon proton donation. A base is a molecular geometry configured to expand and redistribute electron density outward upon proton acceptance. Acid strength and base strength are measures of geometry stability — how energetically favorable the post-reaction charge geometry is relative to the pre-reaction state. pH is reframed as geometry pressure: a quantitative measure of how strongly the solution environment drives molecules into proton-donated or proton-accepted geometry states. Conjugate acid-base pairs are the same molecular framework occupying two different geometry states. Anomalies in acid-base behavior — including the unusual weakness of HF, the kinetics of carbonic acid formation, and the geometry-blocking effects underlying steric inhibition of bases — are explained as geometry reorganization costs, not exceptions to the rules. Solvent effects, including the leveling effect and the enhanced reactivity of bases in

aprotic media, are understood as solvent-driven geometry rewriting of effective charge distributions. The framework also addresses applied domains: titration as geometry restoration, buffers as geometry stabilizers, protein denaturation as catastrophic geometry collapse, and electrochemical processes as interfacial geometry mismatch events. No new theory is introduced. This paper reorganizes existing, well-established physical chemistry into a mechanistic sequence that begins with charge distribution and derives all observable acid-base behavior from it.

1. Acids and Bases as Geometry Behaviors

Start with electrons. Electrons in a molecule are not static — they are constantly pulled by the nuclear charges of every atom in the molecule. The pull each nucleus exerts depends on two things: the magnitude of its nuclear charge (atomic number) and how far away the electron is from that nucleus. When two atoms of different elements share electrons in a covalent bond, those electrons do not sit at the midpoint. They shift toward the atom with the higher effective nuclear pull. This shift is not a metaphor and it is not subtle. It is a physical displacement of electron density in three-dimensional space. That displacement — the spatial asymmetry of where electrons are located in a molecule — is **charge-distribution geometry**.

Every molecule has a unique charge-distribution geometry. Two molecules with different atoms or arrangements will distribute their electrons differently, and that difference determines how they interact with every other molecule they encounter. When a molecule approaches another molecule, what physically happens first is that their charge geometries encounter each other. Regions of high electron density from one molecule approach regions of low or high electron density from the other. What follows — whether a proton transfers, whether a bond forms, whether a complex assembles — is the direct consequence of that geometric encounter.

An **acid** is a molecule whose charge-distribution geometry is configured so that, when it encounters an appropriate partner, its charge compresses. The proton — which carries essentially no electron density of its own, being a bare nucleus — moves toward the partner, and the charge geometry of the remaining molecular framework collapses inward. The molecule before proton transfer has a charge geometry that spans its bonds and includes the proton's contribution. The molecule after proton transfer — the conjugate base — has a charge geometry that has reorganized around the now-absent proton. That reorganization, that inward compression, is what an acid does. It does not "give up" a proton in some vague thermodynamic sense. Its charge geometry distorts to a more compact state when the proton leaves.

A **base** is a molecule whose charge-distribution geometry is configured so that, when it encounters an appropriate partner, it expands to accommodate incoming charge. The lone pair on the base — a region of concentrated negative charge — reaches outward toward an electron-poor center. When a proton arrives, it does not simply "attach." It inserts into the charge geometry of the base, drawing electron density from the base's lone pair toward itself and reorganizing the entire electron density distribution of the molecule into a new, higher-energy (but equilibrium-stable) configuration. The base's geometry opens to accept the proton. This expansion event — the reorganization of charge outward to accommodate incoming positive charge — is what a base does.

These two behaviors — acid as charge compression, base as charge expansion — are not categories into which we sort substances. They are descriptions of what the charge distribution is doing during a reaction. A molecule is not permanently an acid or a base. It is in a geometry state that, given a particular partner and environment, will either compress or expand. Water is the canonical example: in the presence of HCl, water acts as a base, expanding its geometry to accept a proton. In the presence of NH₃, water acts as an acid, compressing its geometry as it donates a proton. The molecule did not change its identity. It responded geometrically to different partners.

The traditional definitional frameworks — **Arrhenius**, **Brønsted-Lowry**, and **Lewis** — are not wrong. They are true at the level of description. Arrhenius correctly observed that acids produce H^+ in water and bases produce OH^- . Brønsted and Lowry correctly identified that acids donate protons and bases accept them. Lewis correctly recognized that the underlying event is electron pair donation and acceptance, which extends the framework to non-proton-transfer reactions. But all three frameworks describe what species participate. None of them explains what the charge is doing. They name the actors without describing the action.

Calling something "an acid" without describing what its electrons are doing is like calling something "fast" without specifying what direction it is moving, what force is propelling it, or what is resisting it. The label carries no predictive mechanism. The charge-distribution geometry framework restores the mechanism. It asks: what does this molecule's electron density map look like, where are the regions of high and low electron density, and what will happen when those regions encounter the electron density map of another molecule? The answer to those questions is the actual chemistry. Everything else — pKa values, acid strength tables, buffer calculations, pH — is a numerical representation of geometry.

Key claim: Acid-base behavior is charge-distribution geometry in action. Arrhenius, Brønsted-Lowry, and Lewis definitions are accurate descriptions of participants, not explanations of mechanism. The mechanism is geometry distortion of electron density.

2. Charge-Distribution Identity

Every molecule has a **charge-distribution fingerprint**. This is not a metaphor. The **molecular electrostatic potential (MEP) surface** is a computed or measurable quantity: it maps the electrostatic potential energy that a test positive charge would experience at every point on the van der Waals surface of a molecule. Where the potential is negative —

red on conventional color-coded MEP maps — electron density is high and the molecule would attract a positive charge. Where the potential is positive — blue — electron density is low and the molecule is electron-deficient at that location. This map is the molecule's charge-distribution identity, and it determines the molecule's chemical behavior before any reaction occurs.

Electron density asymmetry is not a side effect of bonding. It is the primary consequence of bonding and the primary determinant of reactivity. When two atoms bond, they do not share electrons equally unless they are identical atoms (as in H_2 or Cl_2). Every other bond is asymmetric. The asymmetry creates a permanent internal charge distribution — a dipole at the bond level — and the sum of all bond dipoles plus the contribution of lone pairs gives the molecule its overall charge-distribution geometry. This geometry is not incidental. It is the molecule's chemical identity from the perspective of any other molecule it will ever encounter.

Acid-active zones are regions of low electron density — positive electrostatic potential — on the MEP surface. These regions are geometrically positioned to interact with incoming electron density from a base. **Base-active zones** are regions of high electron density — negative electrostatic potential — on the MEP surface. These regions are positioned to donate electron density to an incoming positive charge center (a proton or a Lewis acid). A molecule that has both types of zones on its surface is **amphoteric**. Amphoteric behavior is not a special category. It is a geometric consequence of having acid-active and base-active regions on the same molecular framework.

Consider HCl . The H-Cl bond is strongly polarized toward chlorine because chlorine's effective nuclear charge far exceeds hydrogen's. The hydrogen end carries a large positive electrostatic potential — it is strongly acid-active. The chlorine end carries a large negative electrostatic potential — it is strongly base-active in principle, though in practice the acid-active H end dominates the chemistry because the H-Cl bond is weak enough and the proton departure geometry stable enough that proton transfer occurs readily. HCl 's charge geometry says, unambiguously: the proton will leave.

Consider water. Oxygen pulls electron density strongly from both hydrogen atoms, creating two hydrogen ends with positive electrostatic potential (acid-active zones) and an oxygen lone-pair region with concentrated negative potential (base-active zone). Water's charge geometry is genuinely amphoteric — it has both zones, neither completely dominating the other. What determines which zone participates in any given reaction is the charge geometry of the molecule water encounters. Meet a stronger acid, and water's base-active oxygen lone pair engages. Meet a stronger base, and water's acid-active hydrogen end engages.

Consider NH_3 . Nitrogen is electronegative but less so than oxygen. The three N–H bonds are polar, with hydrogen carrying slight positive potential (weakly acid-active), but the dominant feature of ammonia's MEP surface is the lone pair on nitrogen: a concentrated, highly negative region that is strongly base-active. NH_3 's charge geometry presents a clear base-active face. The molecule will act as a base almost every time it encounters an appropriate partner, because its acid-active zones (the slightly positive H's) are weak and the base-active zone (the N lone pair) is strong. The geometry predicts the behavior before a single molecule reacts.

The practical implication of this framing is significant: you do not need a reaction to predict acid-base behavior. You need the electron density map. Computational chemistry routinely generates MEP surfaces before any experimental work is done, and those maps predict exactly which end of a molecule will donate, which will accept, and what the relative strengths of those interactions will be. The charge-distribution identity is primary. The reaction is the consequence.

Amphoteric molecules — water, amino acids, bicarbonate, HSO_4^- — are simply molecules whose MEP surface has both acid-active and base-active zones of comparable strength. The geometry does not favor one over the other absolutely. Which zone engages depends entirely on the charge geometry pressure of the environment, which is why the same molecule can act as an acid in one context and a base in another. This is not inconsistency. This is geometry responding to geometry.

Molecule	Acid-Active Zone	Base-Active Zone	Dominant Behavior
HCl	H (strongly positive)	Cl (moderately negative)	Strong acid (proton donor)
H ₂ O	H (moderately positive)	O lone pairs (moderately negative)	Amphoteric
NH ₃	H (weakly positive)	N lone pair (strongly negative)	Moderate base
HF	H (strongly positive)	F (very strongly negative)	Weak acid / strong base conjugate
BF ₃	B (strongly positive, empty orbital)	None significant	Lewis acid

3. Proton-Driven Geometry Distortion

A proton is not a typical chemical species. It has no electrons. It is a bare nucleus — a single positive charge of negligible physical size (radius approximately 0.87 femtometers, compared to atomic radii measured in picometers). When a proton is bonded within a molecule, it contributes its nuclear charge to the molecule's overall electrostatic environment, pulling electron density toward itself and distorting the charge geometry of every nearby bond. When it leaves, that pulling force disappears. The molecule's electron density snaps back — or more precisely, reorganizes into the new lowest-energy configuration available to the now-proton-free framework.

Proton transfer is therefore not just bond breaking and bond forming. It is a simultaneous, coordinated **geometry distortion event** across two molecular frameworks. When HCl donates its proton to water, the following happens: the H-Cl bond breaks, and the Cl does not simply "lose H." The entire electron density of the HCl molecule — which was previously spread across the H-Cl bond, with substantial density concentrated at Cl but with a significant contribution at H — collapses inward toward chloride. The proton's positive charge was previously pulling electron density outward from Cl toward H. Its

departure removes that outward pull. Cl^- forms with a spherically symmetric, highly compressed electron cloud. The geometry has gone from an asymmetric, bond-polarized distribution to a perfectly symmetric, concentrated shell.

Simultaneously, on the water side: the incoming proton inserts into the base-active zone of water — the lone pair on oxygen. One of oxygen's two lone pairs commits to forming the new O–H bond. This is not a passive event. The geometry of the water molecule reorganizes. The O–H bond angles change. The two remaining lone pairs on oxygen are now in different geometric positions relative to the new bond. The result is **hydronium** (H_3O^+): a trigonal pyramidal structure with the oxygen at the apex, three O–H bonds arranged below, and a much-reduced lone-pair region on the oxygen (now carrying one lone pair instead of two). The positive charge is formally on oxygen, but it is delocalized across the three equivalent O–H bonds. The charge geometry has gone from a bent, two-lone-pair configuration with moderate negative potential at O to a pyramidal, one-lone-pair configuration with positive potential distributed across the hydrogen positions.

These two geometry changes — the compression of HCl into Cl^- and the expansion/reorganization of H_2O into H_3O^+ — happen together. They are not sequential events separated by a transition state in which the proton floats free. In aqueous solution, the proton is never truly free. It is always associated with water molecules, moving through the hydrogen-bond network via the **Grotthuss mechanism**: sequential proton hopping along pre-existing hydrogen bonds rather than physical diffusion of a proton through solution. Each hop is itself a local geometry distortion event — one O–H bond breaks, one O–H bond forms, and two oxygen frameworks reorganize around the event. The proton transfer that began with HCl and water is therefore not a particle moving from point A to point B. It is a wave of geometry distortion propagating through the charge distribution of the solution.

Key claim: Proton transfer is a geometry distortion event, not a particle-transport event. The conjugate base (Cl^-) and conjugate acid (H_3O^+) represent two new charge geometries that did not exist before the transfer. The proton is the trigger, not the substance of the reaction.

This has direct consequences for how we understand reaction rates. The rate at which a proton transfers between two oxygen atoms in solution is extraordinarily fast — near diffusion-controlled, on the order of $10^{11} \text{ M}^{-1}\text{s}^{-1}$ (Eigen, 1964). This speed reflects the fact that the geometry distortion required — reorganizing oxygen lone pairs to accommodate or release a proton — is minimal. The pre-existing hydrogen-bond geometry of water pre-organizes the system for transfer. The geometry is already partially distorted before the chemical event occurs. Between a nitrogen base and an oxygen acid, the rate is slower because the geometry reorganization is more extensive — the N lone pair geometry and the O-H bond geometry are less pre-organized relative to each other. Proton transfer rate is geometry reorganization rate.

The geometry of the **conjugate base** — whatever molecule results from acid proton donation — is always more charge-compressed than its parent acid. This compression is not accidental. It is what defines the conjugate base: a framework that has lost the outward electrostatic pull of a proton and whose electrons have reorganized inward. The geometry of the **conjugate acid** — whatever molecule results from base proton acceptance — is always more charge-distributed than its parent base. The incoming proton draws electron density outward from the formerly concentrated lone pair, spreading it across a new bond. Both geometry changes are the actual chemical event. The bond-breaking and bond-forming are the mechanisms of those geometry changes, not the other way around.

4. Acid Strength as Compression Stability

Acid strength measures one thing: how much the proton-donated (conjugate base) geometry is energetically preferred over the proton-retained (acid) geometry, in a given environment. That is all pKa measures. A low pKa means the geometry strongly prefers the proton-donated state. A high pKa means it strongly prefers the proton-retained state. Every discussion of acid strength — resonance, induction, polarizability, hybridization effects — is a discussion of what determines that geometric preference.

The question is never "how much does the acid want to give up its proton?" That framing assigns motivation to a molecule. The question is: *how stable is the charge geometry of the conjugate base that results from proton donation?* Stability here means lowest energy. The conjugate base geometry with the lowest energy relative to the starting acid geometry will form most readily, which means the acid that produces it is the strongest.

Consider HCl versus HF. HCl is a strong acid ($\text{pK}_a \approx -7$). HF is a weak acid ($\text{pK}_a \approx 3.2$). Both are hydrogen halides. Both have strongly polarized H-X bonds. Why the enormous difference? Because the conjugate base geometries — Cl^- versus F^- — are radically different in stability. Cl^- has a large atomic radius. Its electron cloud is **polarizable**: the 18 electrons can spread out across a large volume, minimizing electron-electron repulsion and distributing the charge over a large surface area. The charge geometry of Cl^- is comfortable. It is stable. F^- , by contrast, is small and extremely electronegative. Nine electrons crowded into a tiny, high-nuclear-charge ion create an intensely concentrated charge geometry. The electron-electron repulsion in F^- is high relative to its size. The negative charge is not comfortably distributed — it is crammed into a small space with a large nuclear charge pulling inward. F^- is geometrically strained. It is an eager proton-grabber precisely because accepting a proton would relieve that geometric strain.

Resonance delocalization is the most powerful conjugate base geometry stabilization mechanism available to organic acids. When acetic acid (CH_3COOH) loses a proton to form

acetate (CH_3COO^-), the negative charge does not sit on one oxygen. The pi system of the carboxylate group allows the electron density to delocalize across both oxygens equally. The charge geometry of acetate is not a compressed, localized spike at one oxygen — it is a symmetric, extended distribution across two identical oxygen atoms and the intervening carbon. This is a far lower energy geometry than a localized negative charge. The two C–O bonds in acetate are equivalent (bond length ≈ 126 pm, intermediate between a single and double bond) — direct structural evidence that the charge geometry has smeared across both positions.

Compare acetic acid ($\text{pK}_a \approx 4.76$) with ethanol ($\text{pK}_a \approx 16$). Ethanol also loses a proton from an O–H bond. But the resulting ethoxide ($\text{CH}_3\text{CH}_2\text{O}^-$) carries its negative charge exclusively on the oxygen. There is no pi system available to delocalize it. The charge geometry of ethoxide is highly localized — a concentrated negative region on a single oxygen. This geometry is far less stable than acetate, which is why ethanol is 10 billion times less acidic than acetic acid. Ten orders of magnitude in acid strength from one structural feature — resonance geometry stabilization.

Inductive effects provide a secondary geometry stabilization mechanism. Electronegative atoms near the charge center of a conjugate base pull electron density away from that center through the sigma bond framework, effectively spreading the charge further from the primary negative site. Trifluoroacetic acid (CF_3COOH , $\text{pK}_a \approx 0$) is enormously more acidic than acetic acid ($\text{pK}_a \approx 4.76$) because the three fluorines continuously pull electron density away from the carboxylate group, reducing the electron density at the charge center and stabilizing the conjugate base geometry by spreading the charge across the molecule. Each fluorine atom acts as a permanent charge-geometry spreader through the inductive pathway.

Hybridization effects on acid strength reflect the geometry of the orbital from which the proton departs. A proton on a carbon bonded via an sp hybrid orbital (as in an alkyne, C–H of acetylene, $\text{pK}_a \approx 25$) leaves from an orbital with 50% s-character — close to the nucleus, high in electron density. An sp^2 carbon–H (as in benzene, $\text{pK}_a \approx 43$) has 33% s-

character. An sp^3 carbon-H (as in methane, $pK_a \approx 48$) has 25% s-character. The more s-character, the closer the electron density is held to the nucleus, the more effectively the conjugate base carbon retains the electron pair, and the more stable its charge geometry. Alkyne C-H bonds are the most acidic carbon-H bonds precisely because their sp orbitals produce a conjugate base carbanion whose charge geometry is closest to the nucleus and most stable.

The unified statement of acid strength: an acid is strong when its conjugate base geometry is stable — when the electron density distributes effectively across the available framework after proton donation, minimizing electron-electron repulsion and maximizing stabilization by nuclei. Every structural factor that affects acid strength — resonance, induction, polarizability, hybridization, solvent — affects it through this single pathway: how stable is the resulting charge geometry?

5. Base Strength as Expansion Capacity

Base strength is the mirror of acid strength, and it operates through the same geometric logic from the opposite direction. A strong base has a charge geometry that: (1) concentrates electron density in an accessible region, (2) can reorganize cleanly around an incoming proton, and (3) produces a conjugate acid geometry that is stable. All three criteria must be met. High electron density alone does not make a strong base if the geometry cannot reorganize, or if the resulting conjugate acid geometry is strained.

OH^- is a strong base because all three criteria are satisfied simultaneously. Hydroxide carries a full negative charge concentrated on a small oxygen atom — it is geometrically intense and electron-dense. When OH^- accepts a proton, the geometry reorganization is minimal and exquisitely clean: two hydrogen atoms and one oxygen in a bent configuration, no lone pairs that are unhappy, no strained bonds. Water is one of the most geometrically stable small molecules that exists. The conjugate acid of OH^- (water) has a

geometry that is strongly preferred. Strong base $\rightarrow$ stable conjugate acid. Geometry is comfortable after the expansion event.

NH_3 is a weaker base than OH^- , and the geometry difference explains why. Nitrogen's lone pair is less electronegatively concentrated than oxygen's. The N lone pair in ammonia exists in an sp^3 hybrid orbital with less electron density than an oxygen lone pair in the same orbital class, because nitrogen's electronegativity is lower (3.04 vs. 3.44 on the Pauling scale) and the orbital energy is accordingly different. When NH_3 accepts a proton to form NH_4^+ , the resulting geometry is a symmetric tetrahedral ammonium ion. This is geometrically stable, but the expansion event is less favorable because the N lone pair's electron density was lower to begin with — less geometric pressure to expand. Additionally, the positive charge in NH_4^+ is spread across four N-H bonds rather than being neutralized entirely (as in H_2O from $\text{OH}^- + \text{H}^+$). NH_4^+ retains a net positive charge. The geometry stabilization of the conjugate acid is real but less complete than water's neutralization. The result: ammonia is a weaker base than hydroxide by many orders of magnitude (pKb of $\text{NH}_3 \approx 4.7$, pKb of OH^- corresponds to water with $\text{pKa} \approx 15.7$).

Steric effects on base strength are pure geometry effects. The lone pair of 2,6-di-tert-butylpyridine is electronically identical to the lone pair of pyridine — the nitrogen's electron density is essentially the same. But the two bulky tert-butyl groups positioned flanking the nitrogen create a geometric barrier that prevents any incoming electrophile or proton from physically reaching the lone pair. The expansion event cannot happen because the geometry blocks the approach. This base is called a **proton sponge** predecessor, and its behavior demonstrates unambiguously that base strength is not just about electron density — it is about geometric accessibility of that density.

Hard and Soft Acids and Bases (HSAB) theory, developed by Pearson (1963, 1968), is the most explicit acknowledgment in classical chemistry that acid-base interactions are geometry-matching events. In HSAB language: **hard acids** are small, highly charged, non-polarizable — they have concentrated, compact charge geometries. **Hard bases** are small, non-polarizable, and hold their electron density tightly — also concentrated charge

geometry. **Soft acids** are large, low-charge, highly polarizable — diffuse, spread-out charge geometry. **Soft bases** are large, polarizable, loosely holding their electron density — also diffuse geometry.

The HSAB matching principle — hard prefers hard, soft prefers soft — is a geometry-matching rule. A hard acid (concentrated charge geometry) and a hard base (concentrated charge geometry) interact through strong, directional electrostatic interactions — the geometries are complementary in size, shape, and charge density distribution. A soft acid (diffuse, polarizable charge geometry) and a soft base (diffuse, polarizable charge geometry) interact through covalent, orbital-overlap-driven interactions — again, the charge geometries are complementary. Mismatches — hard acid with soft base, or soft acid with hard base — produce weaker interactions because the charge geometry of one partner cannot effectively engage with the charge geometry of the other. HSAB is not a separate theory layered on top of acid-base chemistry. It is the recognition that charge-distribution geometry matching determines interaction strength.

Base	Charge Geometry	Conjugate Acid Geometry	Base Strength (approx. pK _b)
OH ⁻	Concentrated negative; small O, single lone pair sector	H ₂ O — neutral, symmetric, stable	~-1.7 (very strong)
NH ₃	Moderate negative; N lone pair sp ³ orbital	NH ₄ ⁺ — tetrahedral, positive, stable	4.74
CH ₃ NH ₂	Moderate-high; N lone pair, alkyl induction	CH ₃ NH ₃ ⁺ — tetrahedral, positive	3.36
Pyridine	Moderate; N lone pair in sp ² plane	Pyridinium — planar, aromatic	8.77
Aniline	Weak; N lone pair delocalized into ring	Anilinium — geometry disturbs ring	9.40

Aniline's weak basicity makes this point precisely. The lone pair on aniline's nitrogen is not fully localized at nitrogen — it is partially delocalized into the aromatic pi system of the benzene ring. The nitrogen's charge geometry is spread out across the ring, not concentrated at the lone pair site. There is less electron density available at the lone pair for the expansion event. The base is weaker because its charge geometry is already partially expanded — already delocalized. Accepting a proton would localize charge back onto nitrogen and disrupt the ring's aromaticity (itself a geometry stabilization). The geometry prefers to stay delocalized. Aniline's pK_b of 9.4 directly reflects that geometric preference.

6. pH as Charge-Geometry Pressure

pH is defined as the negative logarithm of the hydronium ion activity in solution. This is a correct and useful operational definition. It is also incomplete as a physical description. pH is more fundamentally a measure of **charge-geometry pressure**: the intensity with which the solution is driving molecules into protonated (compressed) or deprotonated (expanded) charge geometry states.

When pH is low — when $[\text{H}_3\text{O}^+]$ is high — the solution contains a large number of hydronium ions pressing against every molecule dissolved in it. Each hydronium ion is a geometry event waiting to happen: it carries a positive charge distributed across its three O–H bonds and presents that charge to every basic site it encounters. A solution with pH 1 has 10^{-1} mol/L of hydronium — ten times the concentration of a pH 2 solution, one hundred times a pH 3 solution. Every basic site in every molecule in that solution is under the constant geometric pressure of proton donation from hydronium. The equilibrium geometry of every acid-base pair in the solution is being pushed toward the protonated state. Low pH means high geometry pressure toward compressed (protonated) states.

When pH is high, the solution is dominated by hydroxide. OH^- is a geometry-expanding agent: it pulls protons from acid sites, converting them from proton-retained to proton-donated charge geometry. High $[\text{OH}^-]$ means high geometry pressure toward expanded

(deprotonated) states. Every acidic proton in every molecule in solution is under pressure to leave. The equilibrium geometry of every acid-base pair is being pushed toward the deprotonated state.

The **Henderson-Hasselbalch equation** — $\text{pH} = \text{pK}_a + \log([A^-]/[HA])$ — is a geometry-state population equation. It describes what fraction of molecules in solution occupy the deprotonated geometry state ($[A^-]$) versus the protonated geometry state ($[HA]$) at any given pH. At $\text{pH} = \text{pK}_a$, the ratio is 1:1 — exactly half the molecules are in each geometry state. The geometry pressure of the solution is balanced against the intrinsic geometry preference of the molecule. The molecule's pK_a is the pH at which neither state is preferred by the environment — the external pressure and the internal geometry preference are equal.

Below $\text{pH} = \text{pK}_a$, the solution's proton pressure exceeds the molecule's preference for the deprotonated state, and the majority of molecules occupy the protonated geometry. Above $\text{pH} = \text{pK}_a$, the solution's deprotonating pressure exceeds the molecule's preference for the protonated state, and the majority occupy the deprotonated geometry. Every unit of pH above or below pK_a represents a 10-fold shift in the population ratio — a direct consequence of the logarithmic scale capturing the exponential relationship between charge-geometry pressure and population distribution.

Buffer systems work because they maintain geometry pressure balance. A buffer consists of a conjugate acid-base pair at concentrations sufficient to absorb proton additions or removals without large shifts in $[H_3O^+]$. When acid is added to a buffer, the added protons are captured by the base-active form (A^-), converting it to the acid form (HA) — a geometry distortion event that consumes the added geometry pressure without releasing it into solution as free hydronium. When base is added, the acid form (HA) donates protons to the base, converting HA to A^- — again absorbing the geometry pressure without releasing it as free hydroxide. The buffer absorbs geometry distortion events and maintains the population ratio, which maintains pH.

Physiological pH regulation is geometry pressure management at the cellular and systemic level. Human blood maintains pH 7.4 ± 0.05 . This narrow window is not arbitrary. It is the geometry-pressure window within which the charge-distribution states of hemoglobin, cytochrome enzymes, membrane ion channels, and metabolic intermediates are all simultaneously in their functional geometry configurations. Hemoglobin's oxygen affinity depends on the protonation state of key histidine residues ($pK_a \approx 6.0$ in the protein environment) — at pH 7.4, these histidines are in the deprotonated state, which enables cooperative oxygen binding. Drop the pH to 7.2 (clinically significant acidemia), and the protonation equilibrium shifts, histidines begin protonating, and hemoglobin's oxygen-binding geometry changes — the Bohr effect. The geometry-pressure shift has functional consequences for every protein in the blood simultaneously. Maintaining pH 7.4 means maintaining the geometry pressure at the exact value where all these molecular populations are in their functional states at once.

7. Conjugate Pairs as Opposite Geometry States

An acid and its conjugate base are not two different molecules. They are the same molecular framework in two different charge-distribution geometry states, separated by one proton and whatever charge that proton's presence or absence implies. Every acid-base reaction is a population-shift event: molecules move from one geometry state to the other in response to the geometry pressure of the environment. The "reaction" is the shift in population, not some singular transformative event.

The geometry-state relationship between an acid and its conjugate base has a precise consequence: the stronger the acid, the weaker the conjugate base, and vice versa — not by coincidence, not by definition, but by geometric necessity. If an acid's geometry strongly prefers the proton-donated state — if the conjugate base geometry is very stable — then the conjugate base has essentially no geometric drive to recapture the proton. Its geometry is already where it wants to be. Strong acid = stable conjugate base geometry = conjugate

base with no enthusiasm for protonation = weak conjugate base. This is the inverse relationship, and it is mechanistically inevitable.

Walk through the major conjugate pairs explicitly, describing the charge geometry of each state:

HCl / Cl⁻. HCl's geometry: strongly polarized H-Cl bond, with H carrying large positive potential and Cl carrying large negative potential. The bond is weak (bond dissociation energy ≈ 432 kJ/mol, lower than H-F). After proton donation, Cl⁻ forms: large, polarizable, spherically symmetric negative charge cloud distributed across 18 electrons in a $3p^6 3s^2$ configuration. The geometry is comfortable, low-stress, stable. Cl⁻ has no geometric incentive to grab a proton back. HCl is a strong acid. Cl⁻ is an extremely weak base.

H₂SO₄ / HSO₄⁻. Sulfuric acid's first proton departs easily ($pK_{a1} \approx -3$). The conjugate base HSO₄⁻ has a geometry in which the negative charge delocalized across four S-O bonds with extensive resonance within the sulfate framework. This is stable geometry. HSO₄⁻ is a weak base — stable geometry, no drive for proton recapture. The second proton of HSO₄⁻ departs less readily ($pK_{a2} \approx 1.99$) because the resulting SO₄²⁻ must accommodate two negative charges in the same framework. Greater charge-charge repulsion. Less stable geometry than HSO₄⁻. Successive deprotonation of a polyprotic acid always gets harder because each additional charge added to the framework increases electrostatic repulsion, destabilizing the conjugate base geometry.

CH₃COOH / CH₃COO⁻. Acetic acid geometry: one C=O and one C-OH, charge concentrated primarily at the carbonyl oxygen and the hydroxyl proton. After proton donation, acetate: symmetric resonance hybrid with charge spread equally across both oxygens. Stable, delocalized geometry. The pKa of 4.76 reflects moderate stability — better than ethoxide (pKa 16) but far worse than Cl⁻ (pKa ≈ -7). Acetate is a weak-to-moderate base, reflecting the partial but not complete stability of the delocalized conjugate base geometry.

NH₄⁺ / NH₃. Ammonium geometry: tetrahedral, four equivalent N-H bonds, positive charge distributed symmetrically. Stable but carries net positive charge. After proton donation (pK_a ≈ 9.25), ammonia: trigonal pyramidal, one lone pair on N concentrated and accessible. NH₃ geometry is a moderate base — the lone pair is available, the expansion event is geometrically accessible, but the electron density is not as high as OH⁻. NH₄⁺ is a weak acid because the tetrahedral geometry is relatively stable and doesn't strongly prefer proton donation.

H₂O / OH⁻. Water geometry: bent, two lone pairs on O, O-H bonds moderately polarized. pK_a ≈ 15.7. After proton donation, hydroxide: single oxygen carrying full negative charge, concentrated and intense. OH⁻ geometry is highly compressed and reactive — a strong base. This means water is a weak acid. And it is: water barely ionizes at all at neutral pH. Its conjugate base (OH⁻) is so geometrically reactive that the equilibrium strongly favors the proton-retained water state.

HF / F⁻. The anomalous pair. HF geometry: strongly polarized bond with H carrying extreme positive potential and F carrying extreme negative potential. The H-F bond is strong (bond dissociation energy ≈ 570 kJ/mol — the strongest H-X bond). After proton donation (pK_a ≈ 3.2), fluoride: tiny, densely negative, geometrically strained as discussed in Section 4. F⁻ is the strongest base among the halide ions, not despite HF being a weak acid, but because of it — the geometry of F⁻ is so unstable (small, concentrated, high nuclear charge) that the equilibrium is pulled back toward HF. These two facts are the same fact expressed from opposite directions.

Polyprotic acids expand this logic into a sequential geometry-state series. H₃PO₄ has three ionizable protons, giving three conjugate pairs and three pK_a values (2.15, 7.20, 12.35). Each successive pK_a is substantially higher because each successive deprotonation forces an additional negative charge onto the phosphate framework, increasing electron-electron repulsion and destabilizing the geometry of the next conjugate base. The molecule resists further compression because it is already partially compressed. The geometry-state

sequence is: $\text{H}_3\text{PO}_4 \rightarrow \text{H}_2\text{PO}_4^- \rightarrow \text{HPO}_4^{2-} \rightarrow \text{PO}_4^{3-}$, with each step costing more geometrically than the last.

8. Anomalies from Geometry Mismatch

Every anomaly in acid-base chemistry is a geometry mismatch event. The framework does not break down at the anomalies. It works precisely there, because geometry mismatch is a real physical cost that shifts equilibria and rates in predictable ways.

The first and most discussed anomaly is the weakness of HF. The hydrogen halides, in order of increasing atomic size and polarizability of the conjugate base anion, show increasing acid strength: HF (pKa 3.2) < HCl (pKa -7) < HBr (pKa -9) < HI (pKa -10). HF is the weakest by an enormous margin, despite fluorine being the most electronegative element and creating the most polarized H-X bond. The naive prediction — most electronegative X pulls hardest on H, most acidic — is wrong. The correct prediction requires looking at the conjugate base geometry, not the acid geometry. F^- is geometrically strained. Its charge density is enormous relative to its size. It exists in a state of intense geometric compression that is partially relieved by accepting a proton from solution. Equilibrium always sits closer to the proton-retained (HF) geometry because releasing F^- into solution forces it into a strained state. This is not HF being "reluctant." This is F^- being so geometrically uncomfortable that it continuously pulls the equilibrium back. The anomaly is not an exception. It is the geometry mismatch cost made numerically visible in the pKa.

The second anomaly is **carbonic acid** and the $\text{CO}_2/\text{H}_2\text{CO}_3$ system. CO_2 dissolves in water to form carbonic acid (H_2CO_3), which is a diprotic acid with $\text{pK}_{\text{a}1} \approx 6.35$ and $\text{pK}_{\text{a}2} \approx 10.33$. But the apparent pKa of the CO_2 /water system is ≈ 6.1 , which is slightly off from the true pKa of H_2CO_3 (≈ 3.6). The discrepancy exists because the reaction $\text{CO}_2 + \text{H}_2\text{O} \rightarrow \text{H}_2\text{CO}_3$ is geometrically slow. CO_2 has a linear, centrosymmetric geometry — the two C=O bonds and the central carbon are in a perfectly symmetric linear arrangement, with zero net dipole moment. Converting CO_2 into H_2CO_3 requires the carbon to go from sp-hybridized linear

geometry to sp^3 -hybridized tetrahedral geometry — a massive geometric reorganization. This reorganization has a significant energy barrier. In the absence of a catalyst, the conversion is slow on the timescale of physiological processes (half-life of hydration reaction ≈ 5 – 10 seconds at body temperature).

This is why **carbonic anhydrase** exists. It is one of the fastest known enzymes (turnover number $\sim 10^6 \text{ s}^{-1}$), and its entire function is to lower the geometric reorganization barrier for the $\text{CO}_2/\text{H}_2\text{CO}_3$ interconversion. The active site contains a Zn^{2+} ion coordinated to three histidines and a hydroxide. The zinc coordinates CO_2 and the hydroxide simultaneously, positioning them so that the geometry reorganization from linear CO_2 to tetrahedral carbonate occurs at the metal center with minimal reorganization cost. The enzyme does not provide energy. It provides a geometric template that pre-organizes the transition state geometry. Enzyme catalysis, at the mechanistic level, is often geometry facilitation — lowering geometric reorganization barriers, not providing thermodynamic driving force.

The third anomaly class involves **Lewis acid geometry reorganization costs**. Boron trifluoride (BF_3) is a Lewis acid — it has an empty p orbital on boron that can accept an electron pair from a Lewis base. But BF_3 is a planar, symmetric molecule (D_{3h} symmetry). The boron has trigonal planar geometry. When a Lewis base approaches and donates an electron pair to the empty boron p orbital, the boron reorganizes from trigonal planar to tetrahedral geometry. This geometry change is a significant geometric reorganization: bond angles change from 120° to approximately 109.5° , the molecule goes from flat to pyramidal, and the three B–F bonds tilt out of the plane. This reorganization costs energy. BF_3 is therefore a weaker Lewis acid than its electron deficiency alone would predict — the geometry reorganization cost partially offsets the thermodynamic gain from the Lewis acid-base interaction.

Compare BF_3 with BCl_3 . BCl_3 is a stronger Lewis acid than BF_3 despite chlorine being less electronegative. The conventional Lewis theory explanation involves pi back-donation: F donates lone pair electron density into the empty B p orbital (p $\rightarrow$ p pi interaction), partially filling the "empty" orbital and reducing BF_3 's electron deficiency. But this is again a

geometry argument: the F lone pairs and the B empty orbital are geometrically positioned for overlap in the planar structure, and this pi donation partially stabilizes the planar geometry and reduces the driving force for Lewis base coordination. Cl is larger, its lone pairs are in larger, more diffuse 3p orbitals that overlap less effectively with boron's 2p orbital, and less pi donation occurs — leaving the boron more electron-hungry and BF_3 more Lewis acidic. Geometry of orbital overlap determines the effective electron deficiency and therefore the effective Lewis acidity.

A fourth anomaly class involves **bifluoride** (HF_2^-): the hydrogen bonding in concentrated HF solutions is so strong that HF dimerizes and higher oligomers form, and F^- associates strongly with HF molecules. This is a geometry-driven association: the intense charge concentration on F^- creates such a strong geometric attraction to the H end of HF that the interaction energy rivals covalent bonds (hydrogen bond energy in $\text{HF}_2^- \approx 163$ kJ/mol, among the strongest hydrogen bonds known). The geometry mismatch of F^- in pure solution is so acute that F^- geometry-seeks any partial positive charge available — including the H of another HF molecule. The anomaly of HF forming bifluoride salts is geometry mismatch driven association, not a special exception to acid-base theory.

These anomalies share a common structure: a geometry reorganization cost, a geometry mismatch strain, or a geometry-driven association that shifts equilibria or rates from naive predictions. None of them breaks the framework. Each is the framework in precise operation at a case where geometric costs are large enough to dominate the chemistry.

9. Solvent-Driven Geometry Rewriting

The solvent is not a passive medium. It is an active geometric participant in every acid-base reaction that occurs within it. The solvent's own charge-distribution geometry — its dielectric constant, its hydrogen-bond donor and acceptor capacity, its polarizability — physically reorganizes around every charged species in solution, changing the effective charge geometry of those species from the outside. Acid strength values measured in water

are not intrinsic molecular properties. They are properties of the acid-in-water system, which is a combined geometric entity.

Water's first-solvation-shell effect demonstrates this directly. When a positive ion forms in water — a proton, a hydronium, a metal cation — water molecules in the first solvation shell immediately orient with their negative oxygen ends facing the positive charge. This orientation is a geometric reorganization: the water molecules align their charge geometry complementarily to the ion's charge geometry, partially neutralizing and distributing the charge across the solvation shell. The effective positive charge of the solvated proton is not +1 in a localized point — it is +1 spread across a cluster of water molecules, with the charge geometry distributed outward into the first and second solvation shells. The H_3O^+ species that we write in chemical equations is the minimum representation — the actual charge geometry extends into the solvation structure (Zundel cation H_5O_2^+ and Eigen cation H_9O_4^+ represent more complete geometric descriptions of the proton in water).

The **leveling effect** is the most striking demonstration of solvent-driven geometry rewriting. Any acid stronger than H_3O^+ — any acid with a pKa below approximately -1.7 in water — is fully leveled to H_3O^+ in aqueous solution. HCl , HBr , HI , HClO_4 , HNO_3 , H_2SO_4 (first ionization) — all of these are "strong acids" in water, but they are not all the same strength. They are all simply stronger than water's proton-handling capacity. Water's charge geometry — with its extensive hydrogen-bond network and strong O lone-pair electron density — reorganizes immediately to capture any proton that any of these acids might donate. Water's geometry is such a powerful proton sink that no acid significantly stronger than H_3O^+ can persist in appreciable concentration. The solvent enforces a ceiling on visible charge asymmetry. Above that ceiling, every excess geometry collapses into H_3O^+ .

To distinguish the true acid strengths of HClO_4 , HCl , HBr , and HI — acids that are all leveled in water — you must use a less basic solvent that does not impose geometry rewriting as aggressively. In acetic acid as solvent (a much weaker base than water), these acids are not fully leveled: $\text{HClO}_4 > \text{HBr} > \text{H}_2\text{SO}_4 > \text{HCl}$ in acetic acid. The ranking becomes

visible because acetic acid's geometry does not reorganize as completely around the proton as water does, leaving the acids' own geometric differences partially unmasked.

Aprotic solvents — DMSO, DMF, acetonitrile, acetone — have no hydrogen-bond-donor capacity. They cannot provide O–H or N–H hydrogen bond donors to stabilize anions through solvation geometry. In water, an anion like F^- or OH^- is partially stabilized by the solvation shell — water molecules orient their H ends toward the anion, distributing the charge outward through hydrogen bonding. This is geometry redistribution by solvation. In DMSO, no such redistribution occurs. The anion's charge geometry is "naked" — unmodified by solvent hydrogen bonding. This has a dramatic effect on reactivity. Bases in DMSO are far more reactive than the same bases in water, because in DMSO the charge geometry of the base is preserved in full concentration, not partially neutralized by solvation geometry.

Reaction rates for $\text{S}_\text{N}2$ reactions, deprotonation reactions, and nucleophilic additions in polar aprotic solvents are orders of magnitude faster than in protic solvents, precisely because the nucleophile/base charge geometry is not dampened by protic solvation. This is why chemists who need powerful bases run reactions in DMSO or DMF: not because these solvents make bases stronger in some abstract sense, but because they stop the solvent from rewriting the base's charge geometry outward into a less reactive distribution. The geometry is preserved.

Superacid systems represent the extreme of solvent-driven geometry rewriting in the acidic direction. Fluorosulfuric acid (FSO_3H), "magic acid" ($\text{FSO}_3\text{H} + \text{SbF}_5$), and related systems have Hammett acidity function values (H_0) below -20 , compared to concentrated H_2SO_4 at $H_0 \approx -12$. These media do not merely provide protons — they impose such an intense positive charge geometry pressure on every dissolved species that they can protonate substrates with essentially no basic sites: hydrocarbons, noble-gas compounds, even molecular hydrogen has been protonated in superacid media. The solvent's geometry pressure is so overwhelming that it forces charge reorganization even on species with no

conventional base-active zones. The geometry of the medium physically distorts the charge geometry of the solute. This is geometry pressure in its most extreme observed form.

The practical takeaway for synthetic and analytical chemists: pKa tables compiled in water are not universal constants. They are water-specific geometry-pressure-balance measurements. When you change the solvent, you change the geometry of every species in it, and acid-base equilibria shift accordingly. Choosing a solvent is choosing a geometry environment. That choice is as chemically consequential as choosing a reagent.

10. Restoring Charge Geometry to Fix Acid-Base Behavior

Acid-base problems — in chemistry, biochemistry, medicine, and materials science — are always geometry problems. Something has the wrong charge-distribution geometry for the function required, or the geometry pressure of the environment is wrong, or a geometry-dependent process is blocked by reorganization cost. Fixing the problem means identifying the geometry mismatch and applying chemistry to shift the population into the required state.

Titration is geometry restoration at the population level. When a strong base is added drop by drop to an acidic solution, each drop converts a fraction of the molecules in solution from their proton-retained geometry state to their proton-donated geometry state. The equivalence point — where the indicator changes or the pH inflection occurs — is the point at which essentially all acid molecules in solution have undergone complete geometry conversion. Every HA has become A⁻. The charge geometry of the solution population has been completely shifted. An acid-base titration curve is a geometry-state population conversion curve plotted against volume of titrant added. The S-shape of the curve reflects the equilibrium geometry preference: near the beginning and end of the titration (far from pKa), the geometry strongly favors one state, so the population shifts readily per proton added. Near the pKa (the midpoint of the curve's steepest section), the population is near

50/50 and is sensitive to small geometry pressure changes — the curve is steep because the population is near the crossover between states.

Buffer design is geometry stabilization by selection of conjugate pair and concentration. To maintain a specific geometry pressure (pH) in a system, choose a conjugate acid-base pair whose pKa is as close as possible to the target pH. This maximizes the buffer capacity — the amount of geometry distortion (proton addition or removal) the buffer can absorb without shifting the population significantly. Then set the concentration ratio of HA to A⁻ using Henderson-Hasselbalch to position the equilibrium exactly at the desired pH. The buffer works because both geometry states are present in sufficient quantity to absorb geometry pressure changes in either direction: A⁻ absorbs incoming protons (geometry compression events), HA absorbs incoming hydroxide (geometry expansion events). A buffer is a geometry shock absorber.

Protein denaturation at extreme pH is a catastrophic geometry collapse. Proteins fold into their functional three-dimensional shapes because of an intricate network of electrostatic interactions, hydrogen bonds, hydrophobic forces, and van der Waals contacts. The electrostatic interactions depend entirely on the protonation state of ionizable residues: histidine (pKa ≈ 6.0–7.0 in protein environments), aspartate (pKa ≈ 3.9), glutamate (pKa ≈ 4.3), lysine (pKa ≈ 10.5), arginine (pKa ≈ 12.5), and the N- and C-termini each have characteristic pKa values that determine whether they carry charge in a given pH environment. These charged residues form salt bridges, interact with metal cofactors, and establish the electrostatic architecture that holds tertiary structure in place.

When pH shifts significantly from the protein's functional range, the geometry-state populations of these residues shift. Histidines, normally deprotonated at pH 7.4 (neutral, non-interacting), begin protonating at pH 6.0 and below (positively charged, now interacting with negative residues or repelling positive residues). Salt bridges that depended on specific charges break when the charge geometry changes. New repulsive interactions form where there were none before. The tertiary structure unravels — not because covalent bonds broke, but because the electrostatic geometry of the folded state

has been destroyed. The protein is not chemically damaged. Its charge geometry is wrong for folding. Renaturation (refolding when the pH is restored) is possible in many cases precisely because the covalent structure is intact and the correct geometry is recoverable.

Metabolic acidosis and alkalosis are systemic geometry pressure dysregulation events. The blood buffer system — the carbonate/bicarbonate/ CO_2 equilibrium — normally maintains plasma pH at 7.40 ± 0.05 . When metabolic processes produce excess acid (lactic acidosis, diabetic ketoacidosis, renal failure with acid retention), the bicarbonate buffer absorbs the initial load ($\text{HCO}_3^- + \text{H}^+ \rightarrow \text{H}_2\text{CO}_3 \rightarrow \text{CO}_2 + \text{H}_2\text{O}$), and the lungs compensate by increasing ventilation to expel CO_2 (a geometry-reducing step — CO_2 removal reduces the acid load). But when the bicarbonate reserve is exhausted or the acid production rate exceeds compensatory capacity, plasma pH falls. At pH 7.2, hemoglobin's oxygen-binding geometry shifts (increased oxygen delivery to tissues — Bohr effect). At pH 7.1, cardiac enzyme geometry begins to fail — arrhythmias occur. At pH 6.8, most enzyme geometries have been pushed outside their functional states and life is not sustainable. The cascade is purely a geometry pressure cascade: one mismatch leads to the next as the population shifts propagate through every pH-sensitive molecular system in the body.

Electrode chemistry and corrosion are charge-geometry mismatch events at material interfaces. Corrosion of iron occurs when the geometry of the metal surface — organized crystalline positive iron nuclei with a delocalized electron sea — encounters the charge geometry of an aqueous acid environment. The geometry mismatch drives charge transfer: electrons from the metal surface reduce H^+ to H_2 (geometry compression — two bare protons capture electrons and become a neutral, bonded molecule), while iron atoms lose electrons and enter solution as Fe^{2+} ions (geometry expansion — neutral metal atoms become bare cations surrounded by a solvation geometry of water molecules). The corrosion process is driven entirely by the geometry mismatch between the metal's electron distribution and the solution's proton/oxygen charge geometry. Corrosion inhibitors work either by physically blocking the interface (geometry separation), by modifying the surface

charge geometry to be less reactive, or by adding species that shift the electrochemical geometry equilibrium to make electron transfer unfavorable.

Battery chemistry is geometry mismatch put to work. A lithium-ion battery maintains a charge-geometry potential difference between the lithium metal/graphite anode and the metal oxide cathode. During discharge, Li^+ ions move from the anode (high electron density, reducing geometry) to the cathode (low electron density, oxidizing geometry), driven by the geometry mismatch. Electrons flow through the external circuit — the usable work output — because the geometry difference between the two electrode materials creates a thermodynamic driving force. Charging the battery restores the geometry mismatch, storing energy as a spatial charge-distribution asymmetry. The battery is literally a stored geometry state difference.

Fixing any acid-base problem — in any of these domains — reduces to the same sequence: identify the geometry state the system is currently in, identify the geometry state required for function, determine what charge-geometry pressure change will shift the population from the current state to the required state, and apply the chemistry. Titration. Buffer design. pH adjustment. Catalyst selection. Solvent choice. Electrode modification. These are all applications of the same principle: charge-distribution geometry can be identified, manipulated, and restored. The chemistry follows the geometry. Fix the geometry; fix the function.

Conclusion

Acid-base chemistry has three established definitional frameworks — Arrhenius, Brønsted-Lowry, and Lewis — each of which correctly describes who participates in acid-base reactions. None of them explains what is physically happening at the charge level. This paper presents a mechanistic reframing: charge-distribution geometry is the physical basis from which all acid-base behavior follows.

The framework rests on a small number of core claims, each mechanistically grounded. Acids are molecular geometries that compress their charge distribution upon proton donation. Bases are molecular geometries that expand their charge distribution upon proton acceptance. Acid strength is compression stability — how stable and low-energy the conjugate base geometry is after the proton leaves. Base strength is expansion capacity — how cleanly and completely the base geometry reorganizes around an incoming proton, and how stable the resulting conjugate acid geometry is. pK_a is the numerical measure of which geometry state is preferred by equilibrium. pH is the geometry pressure of the solution — the intensity with which the proton activity of the medium drives molecules into proton-retained or proton-donated geometry states. Conjugate acid-base pairs are the same molecular framework occupying two different geometry states, separated by one proton.

Anomalies — HF's weakness, carbonic acid's kinetic sluggishness, BF_3 's unexpected Lewis acidity deficit — are geometry reorganization costs and geometry mismatch strains made visible in experimental data. They are not exceptions to the framework. They are the framework at its most precise, operating where geometric costs are large enough to dominate the chemistry. Solvents rewrite effective geometry: water's protic hydrogen-bond network redistributes and dampens charge geometries, aprotic solvents preserve them, and superacid media impose geometry pressure so intense they override the baseline charge geometry of the solute entirely.

Applied to real systems — titration, buffers, protein function, blood pH regulation, electrode chemistry — the framework reduces every intervention to the same operation: identify the geometry mismatch, determine the required geometry state, and apply the chemistry needed to shift the population into that state. Titration is population-level geometry conversion. Buffer design is geometry stabilization by conjugate pair selection. Protein denaturation is catastrophic geometry collapse when pH -driven population shifts destroy electrostatic architecture. Corrosion and battery chemistry are charge-geometry mismatch events harnessed or resisted at material interfaces.

This is not a new theory. Every physical fact invoked here — molecular electrostatic potential surfaces, resonance stabilization, induction, solvation geometry, the Grotthuss mechanism, HSAB theory, the Henderson-Hasselbalch equation — exists in the established literature of physical chemistry, physical organic chemistry, and electrochemistry. What this paper proposes is a reorganization: start with charge distribution, derive geometry, derive geometry behavior, derive everything else from that. Put the mechanism first. Stop describing acid-base chemistry in terms of categories and start describing it in terms of what the electrons are doing.

Chemists, biochemists, materials scientists, and clinicians who operate with a mechanism-first mental model make better predictions and more rational interventions. They ask the right question first: not "what kind of acid is this?" but "what is this molecule's charge geometry, how does it compare with what it is encountering, and which state does the equilibrium prefer?" Everything else follows from that. The goal of this framework is to make that question — and its answers — the natural starting point for thinking about acid-base chemistry.

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